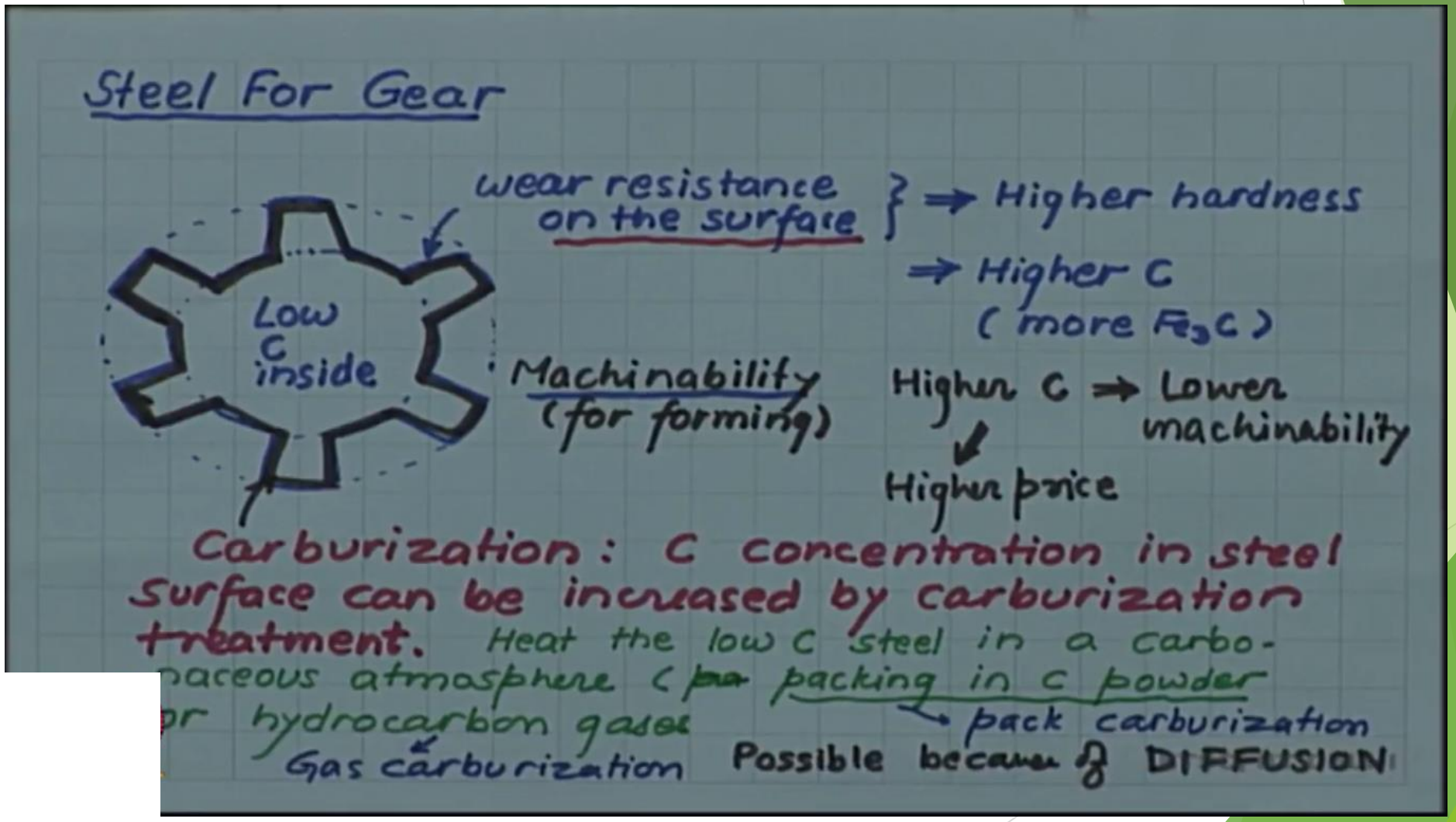
LASC MED201

Class 7 & 8

Chapter 5

- Diffusion
  - Fick's Law (I & II)
- Phase Transformations
  - TTT Diagrams

Diffusion

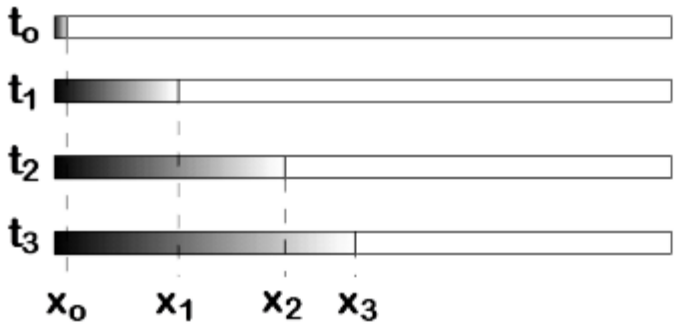

Steel For Gear
wear resistance on the surface
Higher hardness
Higher C
(more Fe_3C)
Low C inside
Machinability (for forming)
Higher C → Lower machinability
Higher price
Carburization : C concentration in steel surface can be increased by carburization treatment. Heat the low C steel in a carbo-
aceous atmosphere (packing in C powder
or hydrocarbon gases
pack carburization
Gas carburization
Possible because of DIFFUSION

What is Diffusion?

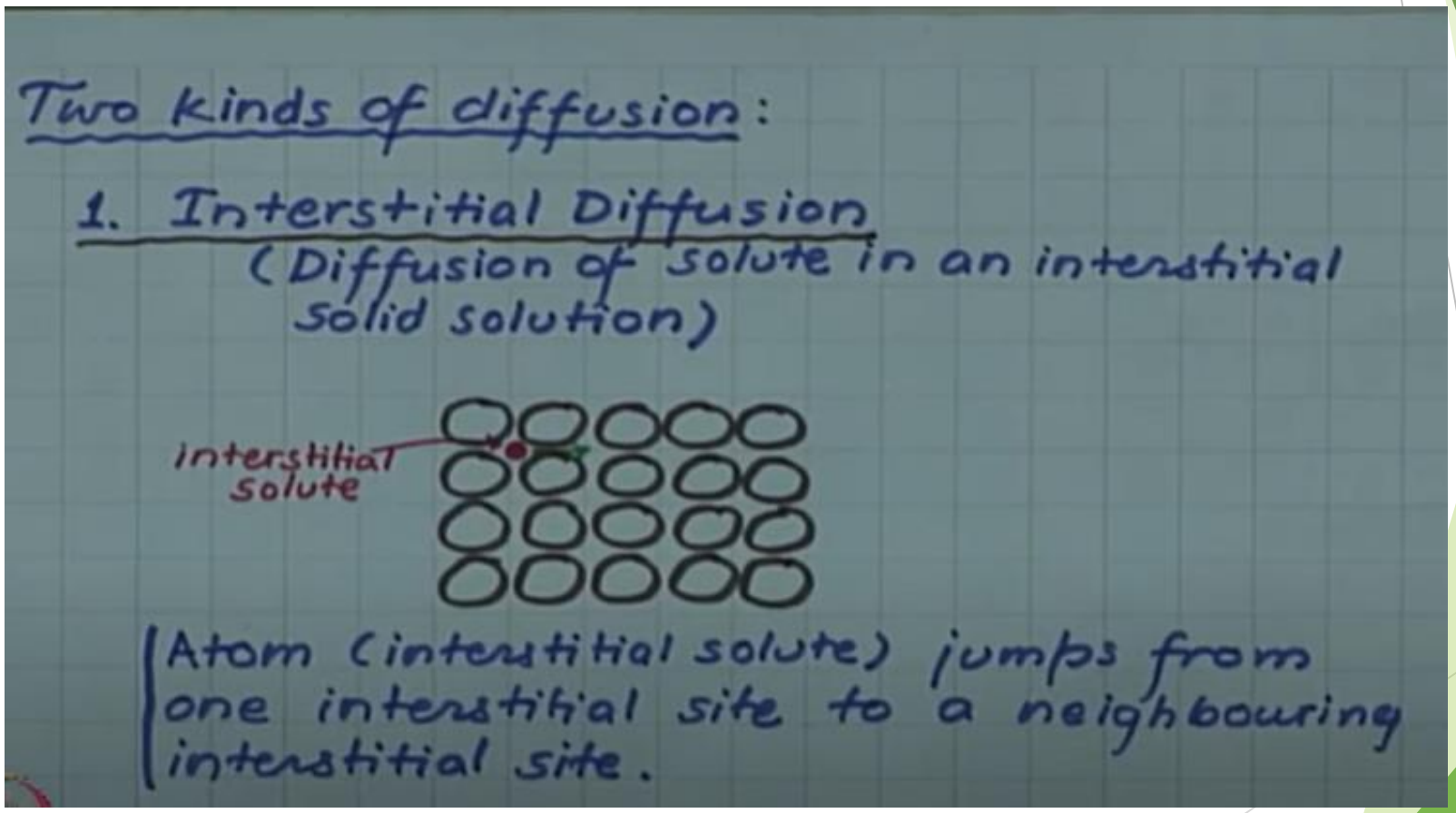
Diffusion is a process of **material transport by atomic motion** under the influence of **thermal energy** & **concentration gradient**

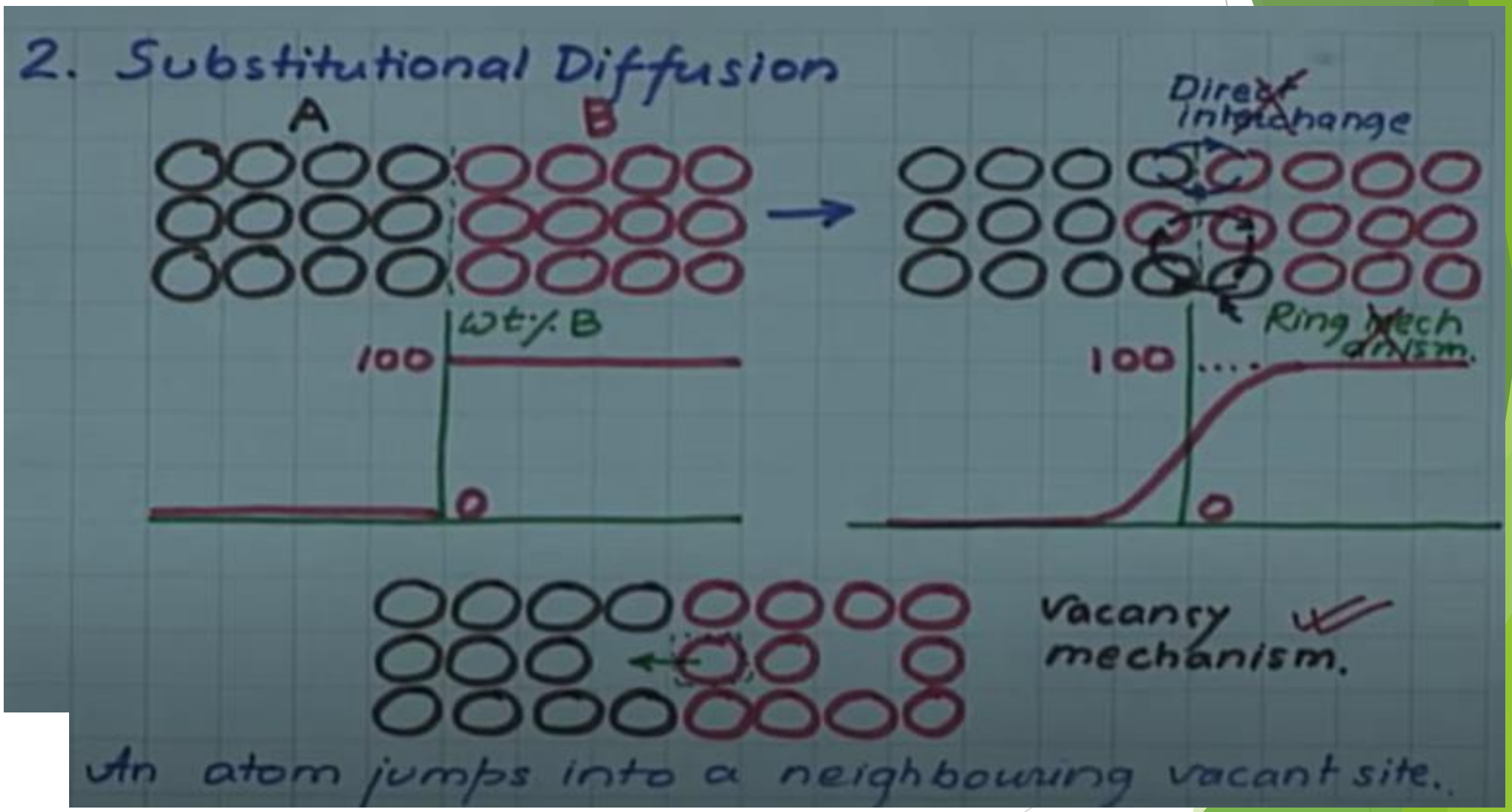
→ *Atoms move from higher to lower concentration region*

- Glass tube filled with water.
- At time t = 0, add some drops of ink to one end of the tube.
- Measure the diffusion distance, x, over some time.



Types of Diffusion: Mechanism




2. Substitutional Diffusion
A
B
Wt% B
100
0
Direct Interchange
Ring Mechanism.
100
0
Vacancy mechanism.
An atom jumps into a neighbouring vacant site.

Comparison

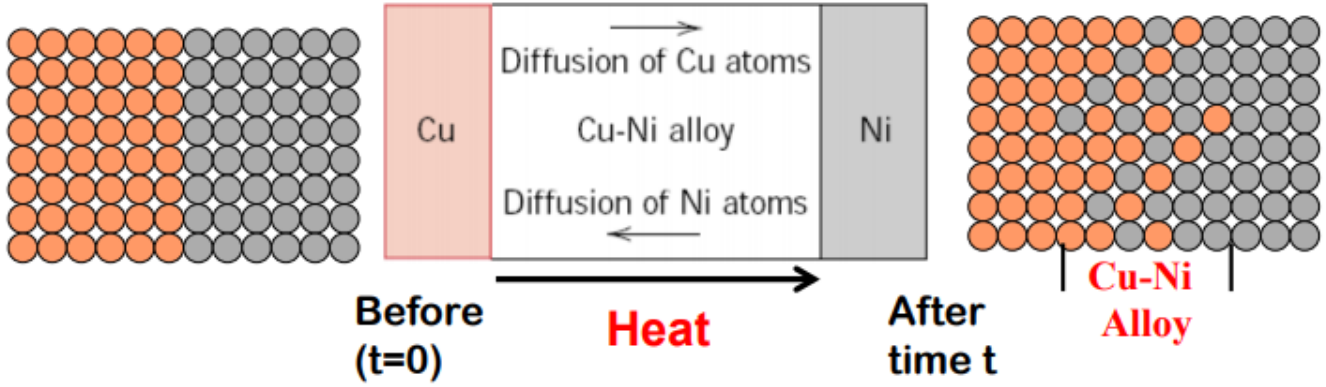
Interstitial diffusion is generally faster than substitutional diffusion.

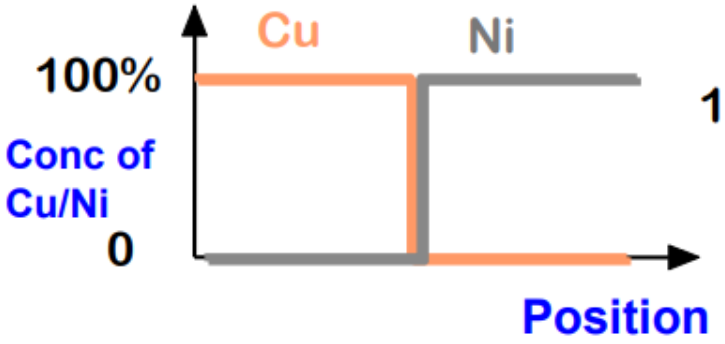
Probability of finding a neighbouring vacant interstitial is much higher than probability of finding a neighbouring vacancy for substitutional diffusion.

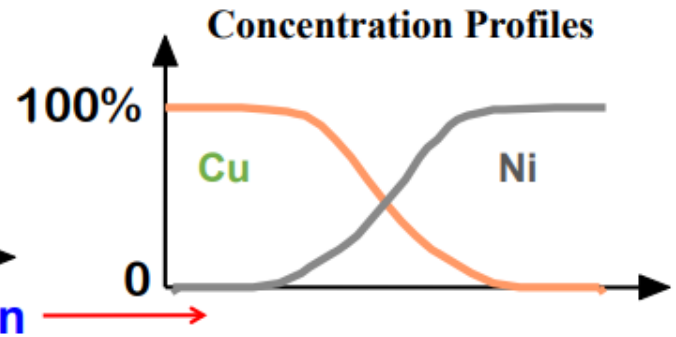
Diffusion Phenomenon

Diffusion: The Phenomena (1)

Inter-diffusion (or impurity diffusion): In an alloy, atoms tend to migrate into another atoms in response to **concentration gradient**

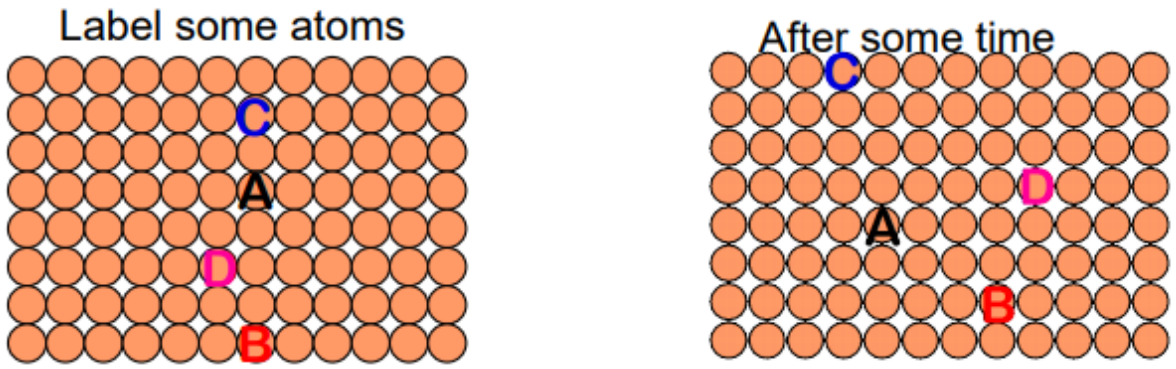






Diffusion: The Phenomena (2)

Self-diffusion: In an one-component material, when atoms that exchange positions are of the **similar type** (pure metals)



Migration of atoms from one position to another

Diffusion Mechanism

For atoms to move, two conditions must be met.

1) There must be an empty adjacent site

2) The atoms must have enough energy to break the bonds with its neighbor atoms- *vibrational energy through temperature increase.*

Fick's Law (I and II)

	Heat Flow	Charge Flow	Mass Flow
1) Flux	Heat Flux q $(J\,m^{-2}s^{-1})$	Charge Flux j (Current density $(A\,m^{-2} \equiv C\,m^{-2}s^{-1})$	Mass Flux j $(kg\,m^{-2}s^{-1},\ mol\,m^{-2}s^{-1})$
2) Gradient	Temperature Gradient $\frac{dT}{dx}$	Electric Potential Gradient $\frac{dV}{dx}$	Concentration Gradient $\frac{dC}{dx}$
3. Law	Fourier's Law of Heat Conduction $q = -K\left(\frac{dT}{dx}\right)$	Ohm's Law $j = -\sigma \frac{dV}{dx}$	Fick's First Law $\boxed{j = -D\frac{dC}{dx}}$ $C \equiv kg\,m^{-3},\ mol\,m^{-3}$
4. Material Property	Thermal conductivity K $(W\,m^{-1}K^{-1})$	Electrical Conductivity σ $(S\,m^{-1} \equiv \Omega^{-1}m^{-1})$	Diffusivity or Diffusion coefficient (m^2s^{-1})

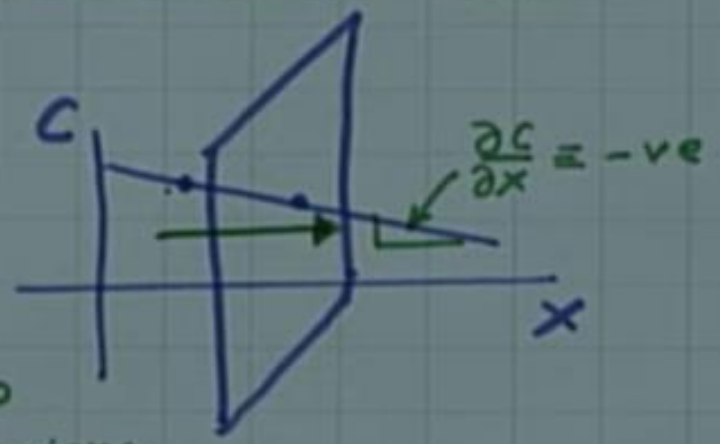
Switch to Sir's slide for laws

FICK'S FIRST LAW

(+ve) +ve −ve

$$\boxed{j = -D\frac{\partial c}{\partial x}}$$

j = Mass Flux



Mass flows from high concentration to low concentration regions.

D ≡ Diffusivity or Diffusion coefficient.

$$[D] = \frac{[J]}{\left[\frac{\partial c}{\partial x}\right]} = \frac{kg\ m^{-2}\ s^{-1}}{\left(\frac{kg/m^3}{m}\right)} = m^2 s^{-1}$$

c in $kg\ m^{-3}$

Diffusivity D :

Highly temperature Dependent

$$D = D_0 \exp\left(-\frac{Q}{RT}\right)$$

Activation Energy

Absolute temperature

Preexponential Factor

Gas Constant

Will recommend to solve numericals of Fick's law from book.

Structure and Diffusion

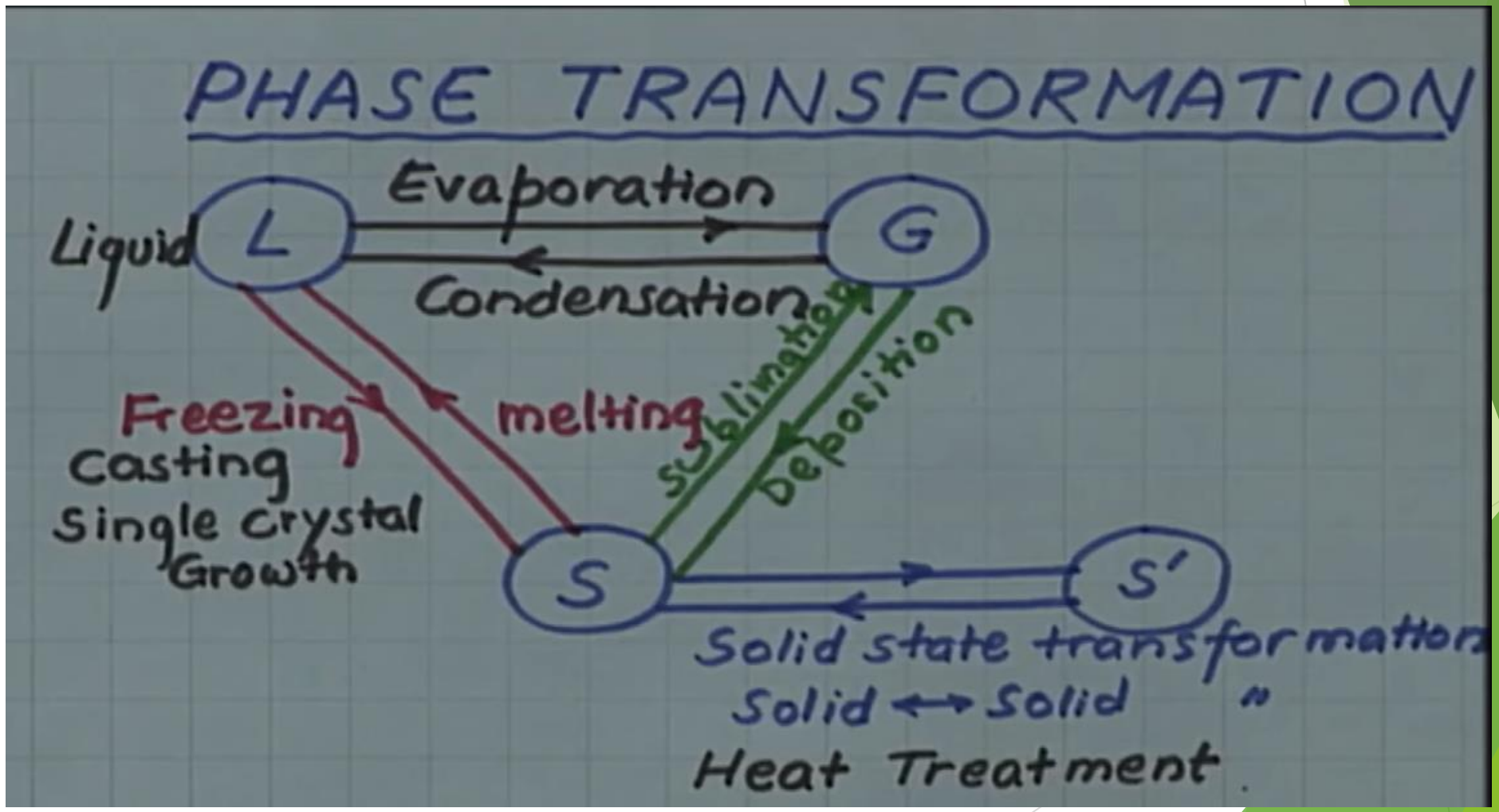
Diffusion FASTER for...

- Open crystal structures
- Lower melting T materials
- Materials w/secondary bonding
- Smaller diffusing atoms
- Lower density materials

Diffusion SLOWER for...

- Close-packed structures
- Higher melting T materials
- Materials w/covalent bonding
- Larger diffusing atoms
- Higher density materials

Phase Transformation



Phase transformations (change of the microstructure) can be divided into three categories:

- Diffusion-dependent with no change in phase composition or number of phases present (e.g. melting, solidification of pure metal, recrystallization, etc.)
- Diffusion-dependent with changes in phase compositions and/or number of phases (e.g. eutectic or eutectoid transformations)
- Diffusionless phase transformation - by cooperative small displacements of all atoms in structure, e.g. martensitic transformation

Driving force for phase transformation

What is the thermodynamic driving force for a phase transformation?

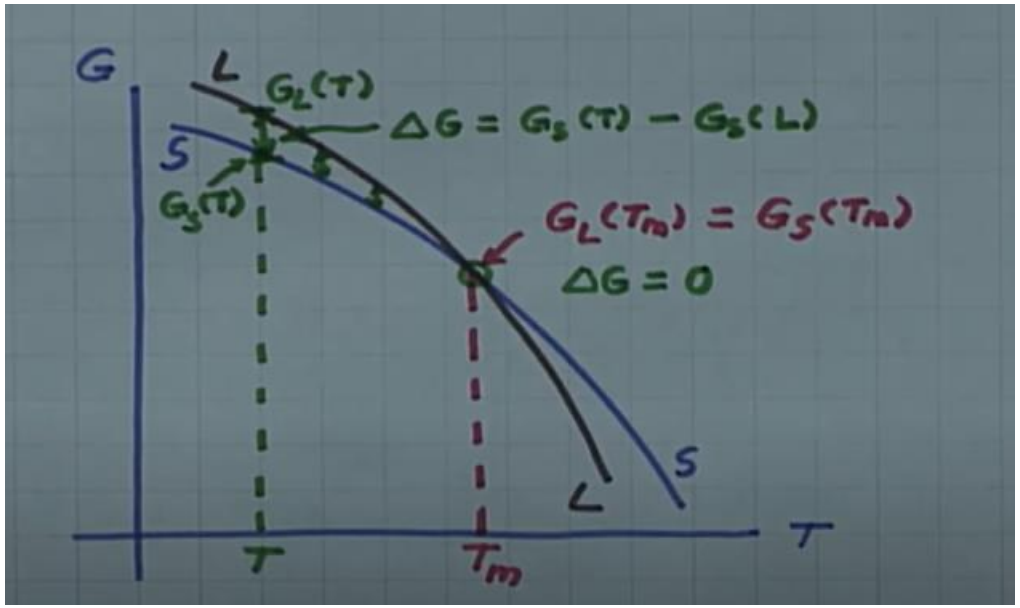
$$\alpha \longrightarrow \beta$$

$$G_\alpha \qquad G_\beta$$

If $\boxed{G_\beta < G_\alpha}$ $\Rightarrow$ Transformation thermodynamically feasible

Driving force $\Delta G = G_\beta - G_\alpha$

Delta G derivation



Estimation of ΔG for freezing

$$\Delta G = G_S - G_L$$

$$\Delta G(T) = G_S(T) - G_L(T)$$

$$\Delta G(T_m) = G_S(T_m) - G_L(T_m)$$

$$= 0$$

$$G_S = H_S - T S_S$$

$$G_L = H_L - T S_L$$

$$\Delta G(T) = \Delta H(T) - T \Delta S(T_m)$$

$$\Delta G(T_m) = \Delta H(T_m) - T_m \Delta S(T_m)$$

$$\Rightarrow \quad 0 = \Delta H(T_m) - T_m \Delta S(T_m)$$

$$\Rightarrow \quad \Delta S(T_m) = \frac{\Delta H(T_m)}{T_m}$$

Latent heat of melting

$$\Rightarrow \quad \Delta S_m = \frac{\Delta H_m}{T_m}$$

$\Delta S_m \equiv \Delta S(T_m)$

$\Delta H_m \equiv \Delta H(T_m)$

Assuming $\Delta H(T)$ and $\Delta S(T)$ to be independent of T:

$$\Delta H(T) = \Delta H(T_m) = \Delta H_m$$

$$\Delta S(T) = \Delta S(T_m) = \Delta S_m$$

$$\Delta G(T) = \Delta H(T) - T \Delta S(T)$$

$$= \Delta H_m - T \Delta S_m$$

$$= \Delta H_m - T\left(\frac{\Delta H_m}{T_m}\right) =$$

Main Formula

$$\Delta G(T) = \Delta H_m - \frac{T}{T_m}\Delta H_m$$

$$= \left(1 - \frac{T}{T_m}\right)\Delta H_m$$

$$= \frac{T_m - T}{T_m}\Delta H_m$$

$$\boxed{\Delta G(T) = \frac{\Delta T}{T_m}\Delta H_m}$$

$$\Delta T = T_m - T$$

$$= \text{undercooling}$$

phase transformation

= nucleation + growth

first formation of a new phase (nucleus)

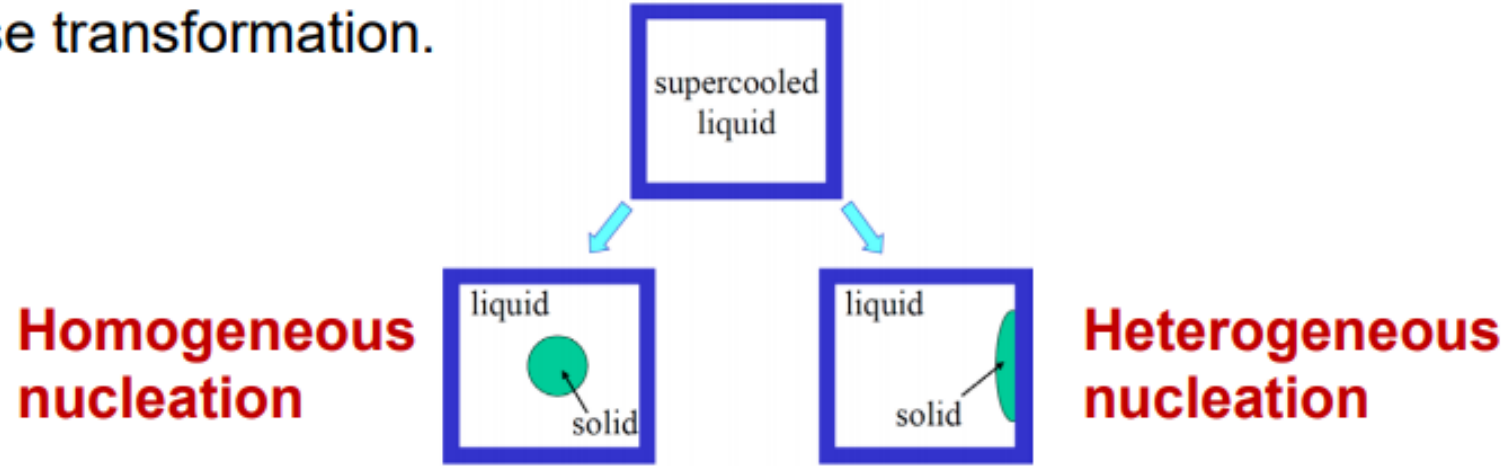
further increase in the size of the nucleus

Nucleation

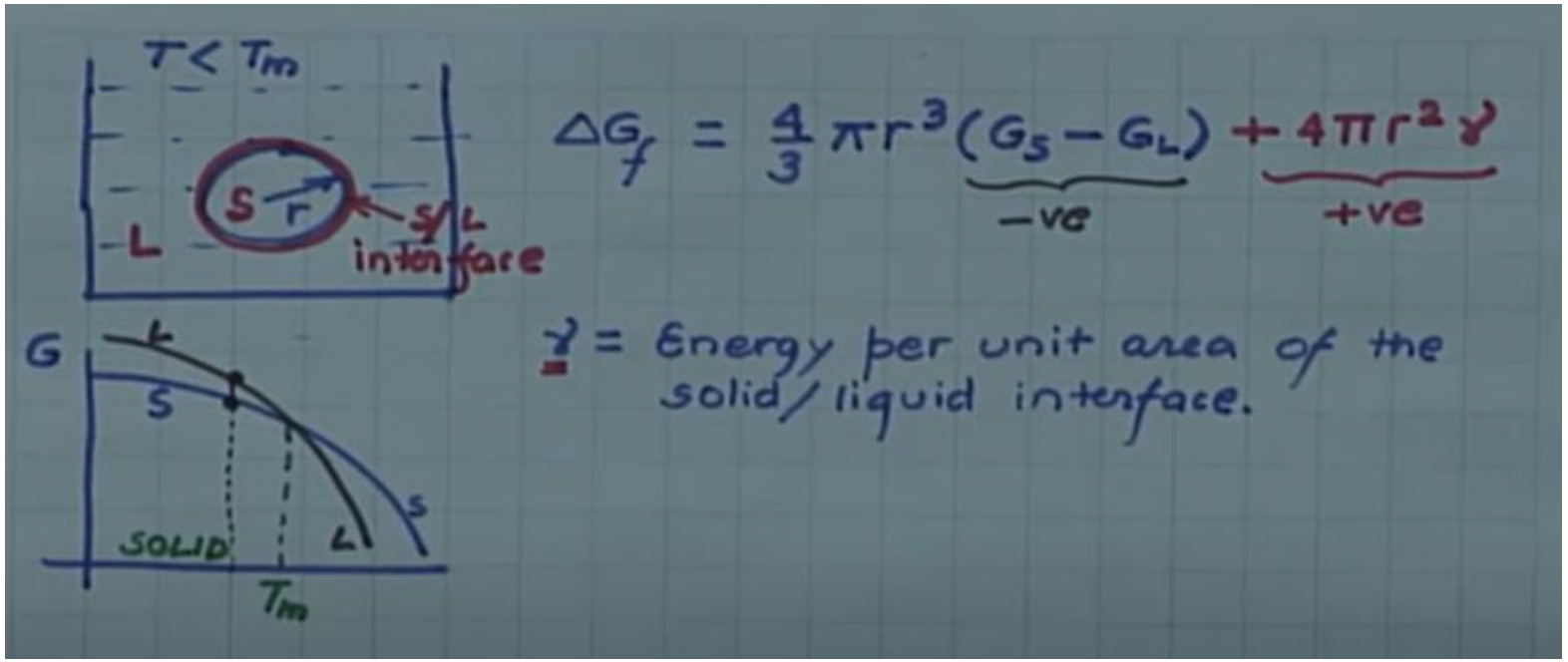
Two types of nucleation – **Homogeneous** and **Heterogeneous**

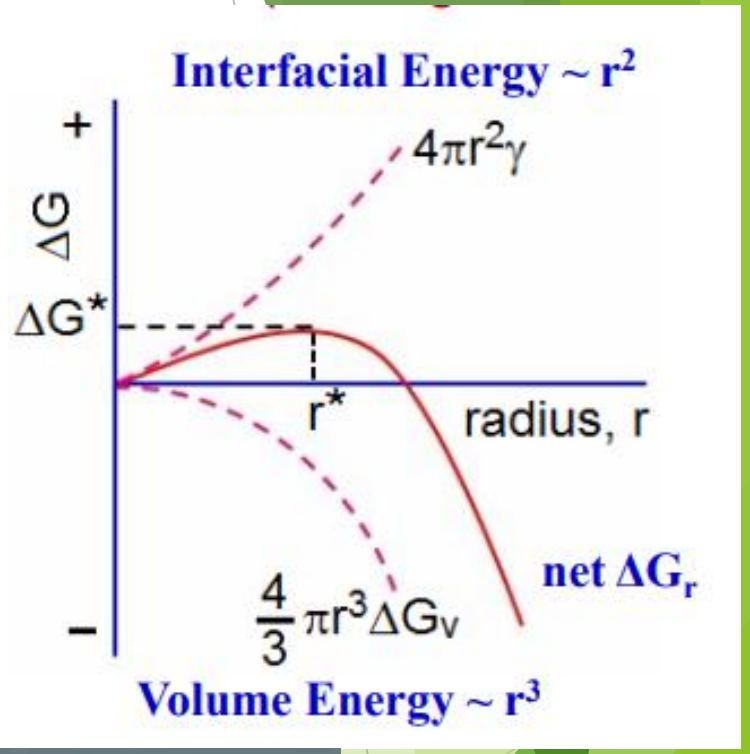
- **Homogeneous** – solid nuclei spontaneously appear within the under-cooled phase completely surrounded by the parent phase.
- **Heterogeneous** – the new phase appears on the walls of the container, at impurity particles, grain boundaries etc.

Let's consider solidification of a liquid phase *under-cooled* below the melting temperature as a simple example of a phase transformation.



Homogenous Nucleation





For $r < r^*$ $\Delta G_f \uparrow$ as $r \uparrow$. ⇒ Growth not possible.

$r > r^*$ $\Delta G_f \downarrow$ as $r \uparrow$ ⇒ Growth is POSSIBLE

r^* = CRITICAL RADIUS FOR NUCLEATION

Critical Radius

There are two contributions to free energy change, volume free energy ΔG_v (free energy difference, $G_s - G_l$) and surface free energy, γ (solid/liquid interfacial energy) due to creation of a new surface.

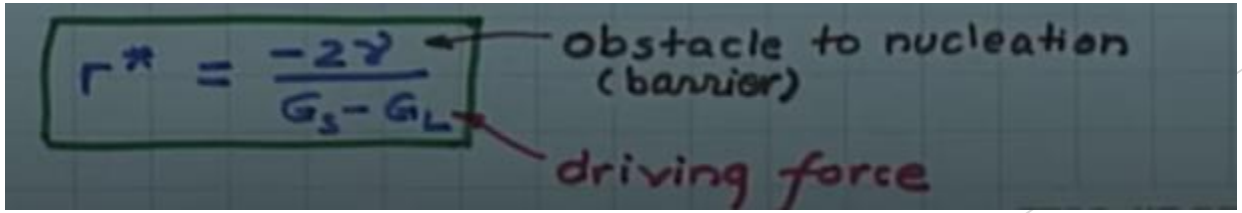
Taking the nucleus as a spherical particle of radius r

$$\Delta G = -4/3\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$

The tiny particle of the solid that forms first will be stable only when it achieves a critical radius (r^*). Below the critical radius it is unstable and is called *embryo*. Stable nuclei, $r > r^* \rightarrow \Delta G$ decrease

- Since this happens at the maximum of the *ΔG vs. r curve*

$$d\Delta G/dr = -4\pi r^2 \Delta G_v + 8\pi r\gamma = 0 \quad \Rightarrow \quad r^* = \frac{2\gamma}{\Delta G_v}$$



Critical r with Undercooling

Latent heat of fusion

$$G_S - G_L = \Delta G_v = \frac{\Delta H_m \Delta T}{T_m} \qquad \Delta T = T - T_m$$

undercooling.

melting point.

$$r^* = -\frac{2\gamma}{G_S - G_L} = -\frac{2\gamma}{\Delta G_v}$$

$$= -\frac{2\gamma T_m}{\Delta H_m \Delta T}$$

Higher undercooling (lower T of solidification).

→ smaller r^*.

Thus $$\Delta G^*_{Hom} = \frac{16\pi\gamma^3}{3(\Delta G_v)^2}$$

ΔG_v is proportional to under-cooling (ΔT) below melting temp. :

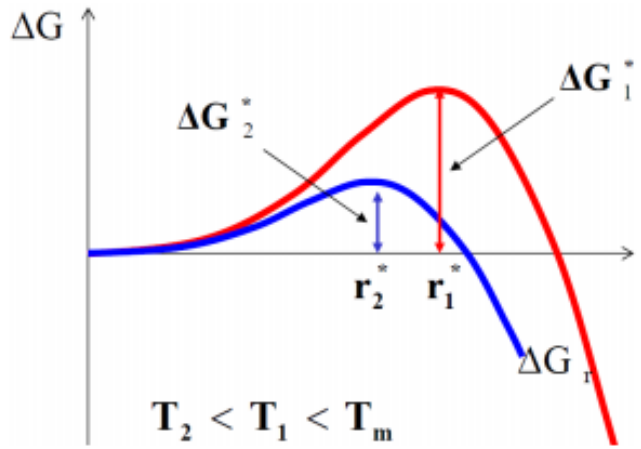
$$\Delta G_V = \frac{\Delta H_m \Delta T}{T_m}$$

H_m is latent heat of melting (or fusion)
$\Delta T = T_m - T$

Therefore

$$r^* = \left(\frac{2\gamma T_m}{\Delta H_m}\right)\frac{1}{\Delta T} \qquad \Delta G^*_{Hom} = \left(\frac{16\pi\gamma^3 T_m^2}{3(\Delta H_m)^2}\right)\frac{1}{(\Delta T)^2}$$

Both r* and G* decrease with increasing under-cooling



ΔG* is energy barrier for formation of a solid nucleus of critical size r*

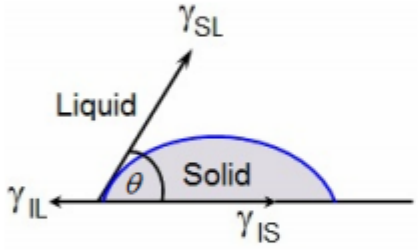
Go through Rate of Homogenous nucleation from sir's slide. Imp Part

Heterogenous Nucleation

Heterogeneous Nucleation

Here, the probability of nucleation is much higher at certain preferred sites such as walls of the container, at impurity particles (inclusions), grain boundaries, etc., compared to rest of the parent phase.

- Example - Solidification of a liquid on an **inclusion surface**



$$\gamma_{IL} = \gamma_{IS} + \gamma_{SL} \cos\theta$$

θ is wetting angle

- With a similar approach it can be shown that

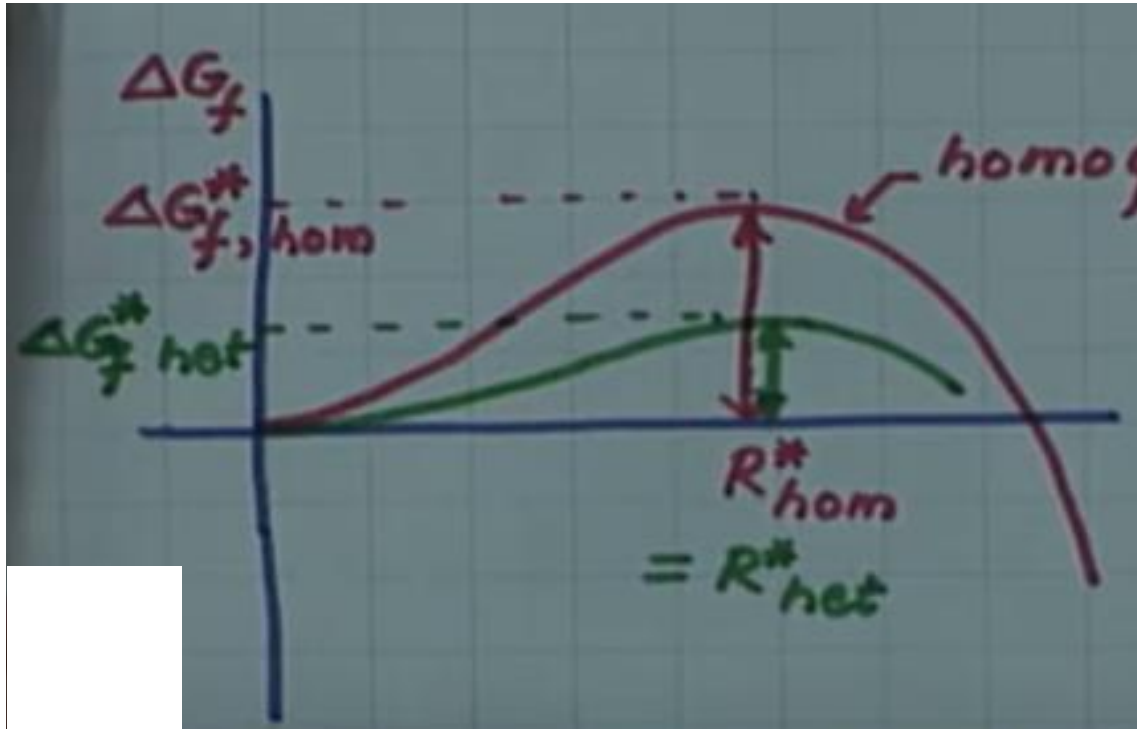
$$r^* = -\frac{2\gamma_{SL}}{\Delta G_v}$$

$$\Delta G^*_{Het} = \frac{16\pi\gamma^3}{3(\Delta G_v)^2} S(\theta) = \Delta G^*_{Hom} S(\theta)$$

$$S(\theta) = \frac{2 - 3\cos\theta + \cos^3\theta}{4}$$

- The small value of θ ensures that the energy barrier (ΔG) is effectively lowered in heterogeneous nucleation.

Homogenous vs Heterogenous Nucleation



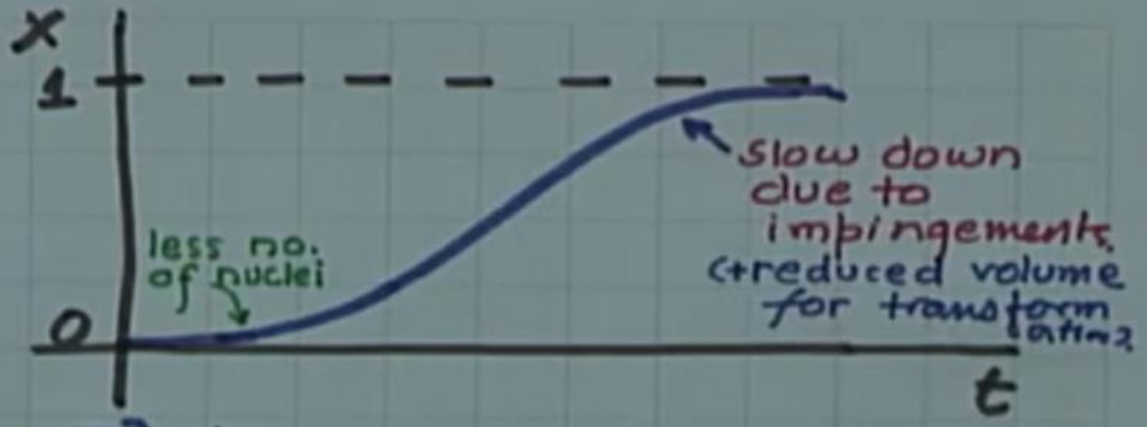
$$\Delta G^*_{f\,het} = \Delta G^*_{f\,hom} \cdot S(\theta)$$

$$V^*_{het} = V^*_{hom} \cdot S(\theta)$$

$$\theta \downarrow \longrightarrow V^*_{het} \downarrow$$
$$\Delta G^*_{het} \downarrow$$

Heterogeneous nucleation is favoured over homogeneous nucleation for smaller contanct angles θ.

Nucleation, Growth and Overall Rate



Nucleation rate, I ($m^{-3}s^{-1}$) : No. of nucleation events per unit volume per second.

Growth rate, $G = \frac{dR}{dt}$ ($m s^{-1}$) : Rate of increase of the size of growing particle (R : particle size)

Overall transformation rate $\frac{dX}{dt}$ (s^{-1}) : Fraction transformed per second.

$X \equiv$ fraction transformed.

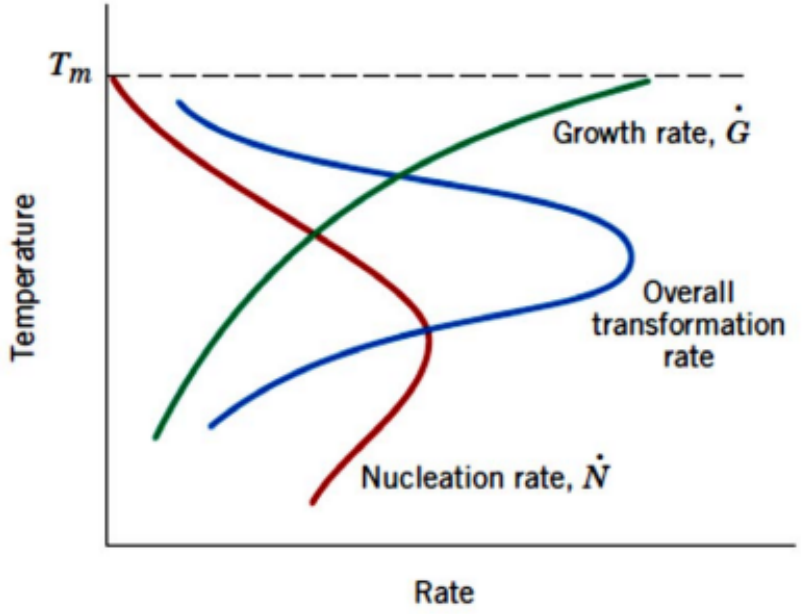
Nucleation and Growth Kinetics

Once the embryo exceeds the critical size r*, the growth of the nucleus starts. Nucleation continues simultaneously.

➢ Nucleation and growth rates are function of temp. Nucleation rate increases with cooling rate and degree of under-cooling ($\Delta T = T_m - T$) while growth rate decreases with ΔT

➢ Total rate of a phase transformation is the product of nucleation rate and growth rates.

- High T (close to T_m): low nucleation and high growth rates
  → coarse microstructure with large grains
- Low T (strong undercooling): high nucleation and low growth rates
  → fine structure with small grains
  (i.e finer grain size)



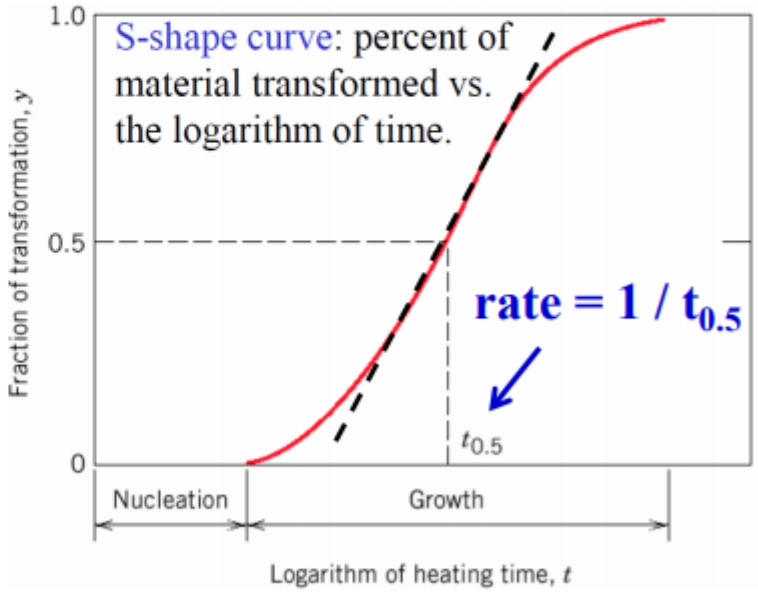
Kinetics

Kinetics of Phase Transformations (solid-solid)

The *time* dependence of solid-state phase transformations at a ***fixed temperature*** is often described in terms of the time dependence of the fraction of transformation (*y*):

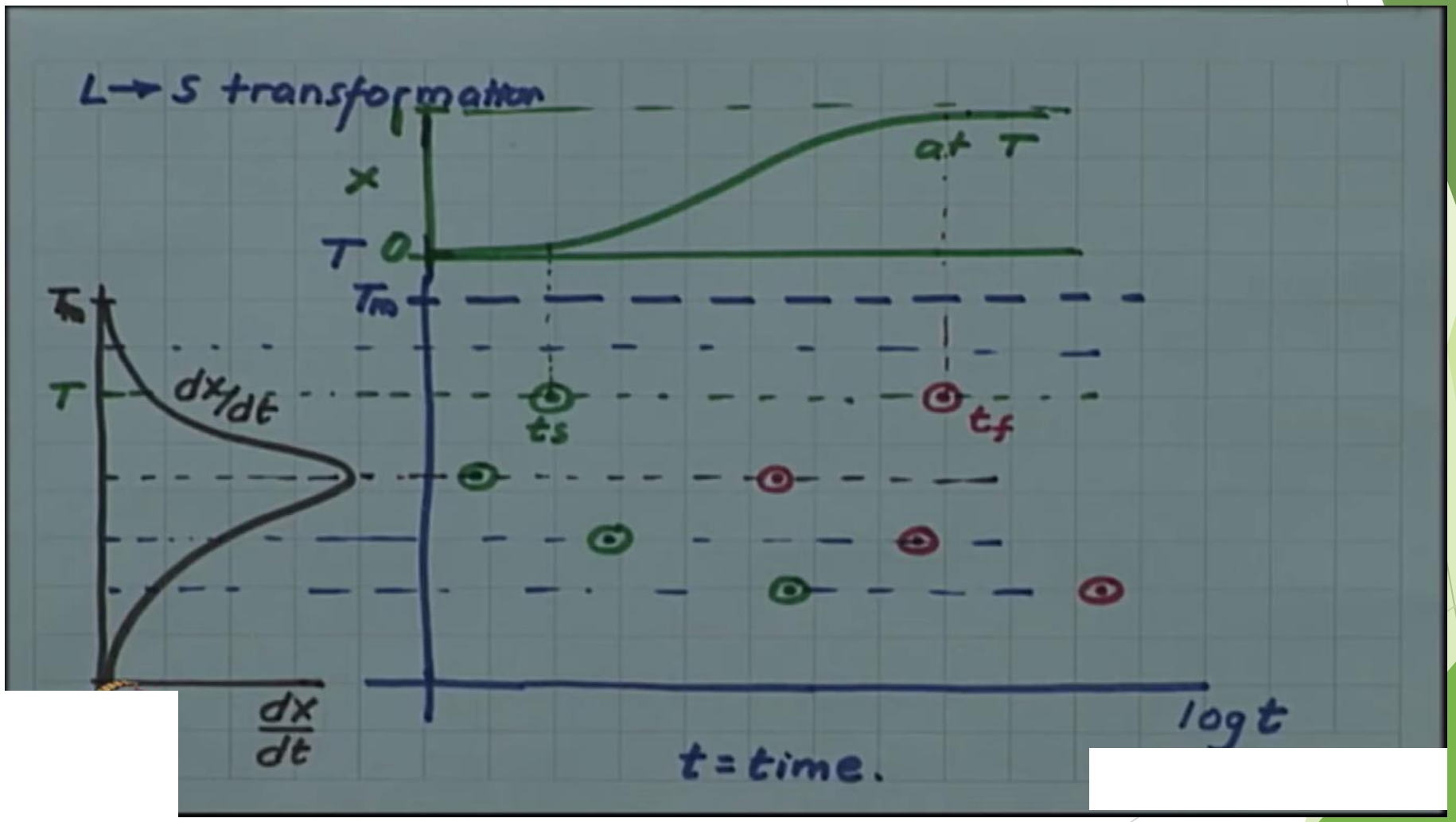
For solid-state transformations displaying the kinetic behavior, the fraction of transformation *y* is a function of time *t* as follows:

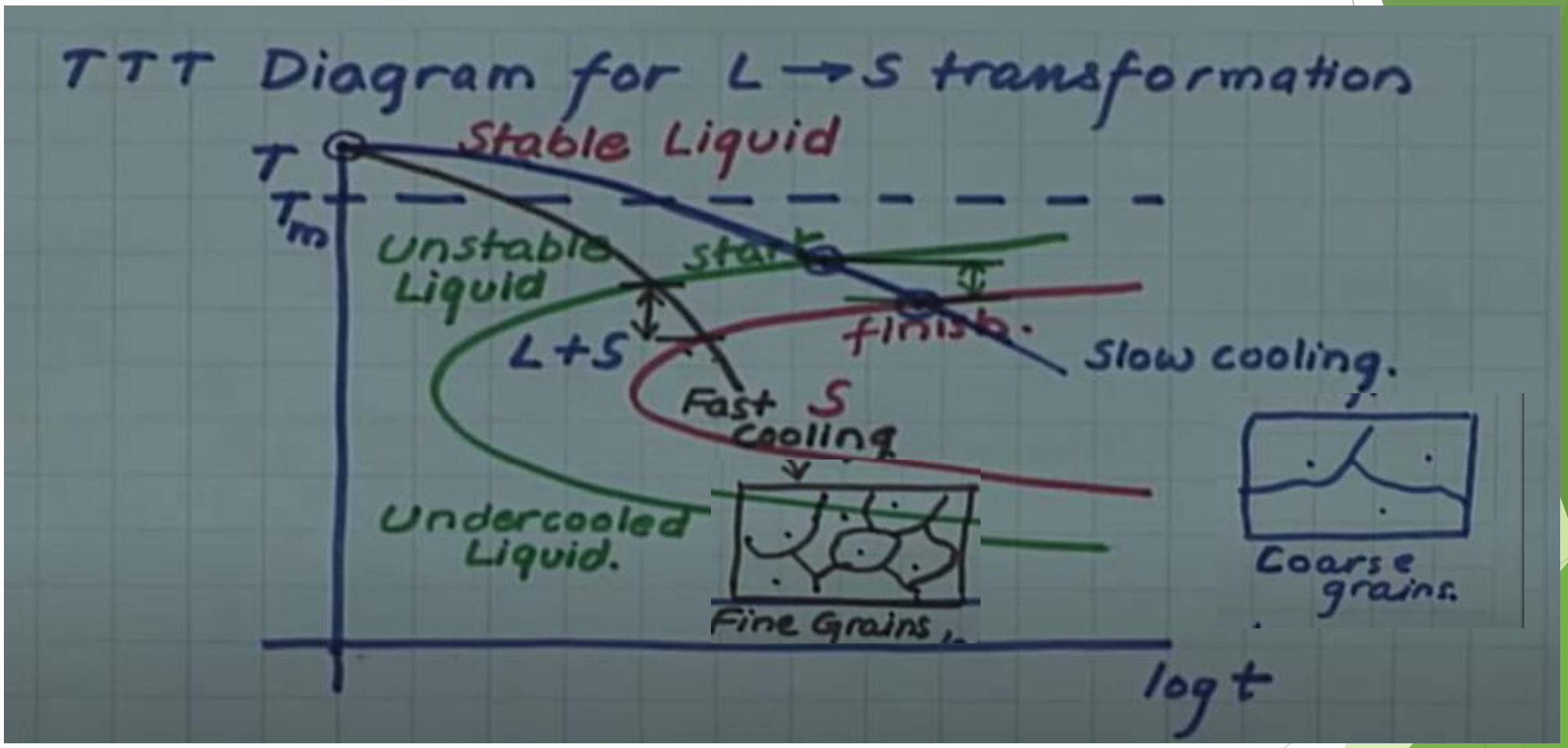
$$y = 1 - \exp(-kt^n)$$



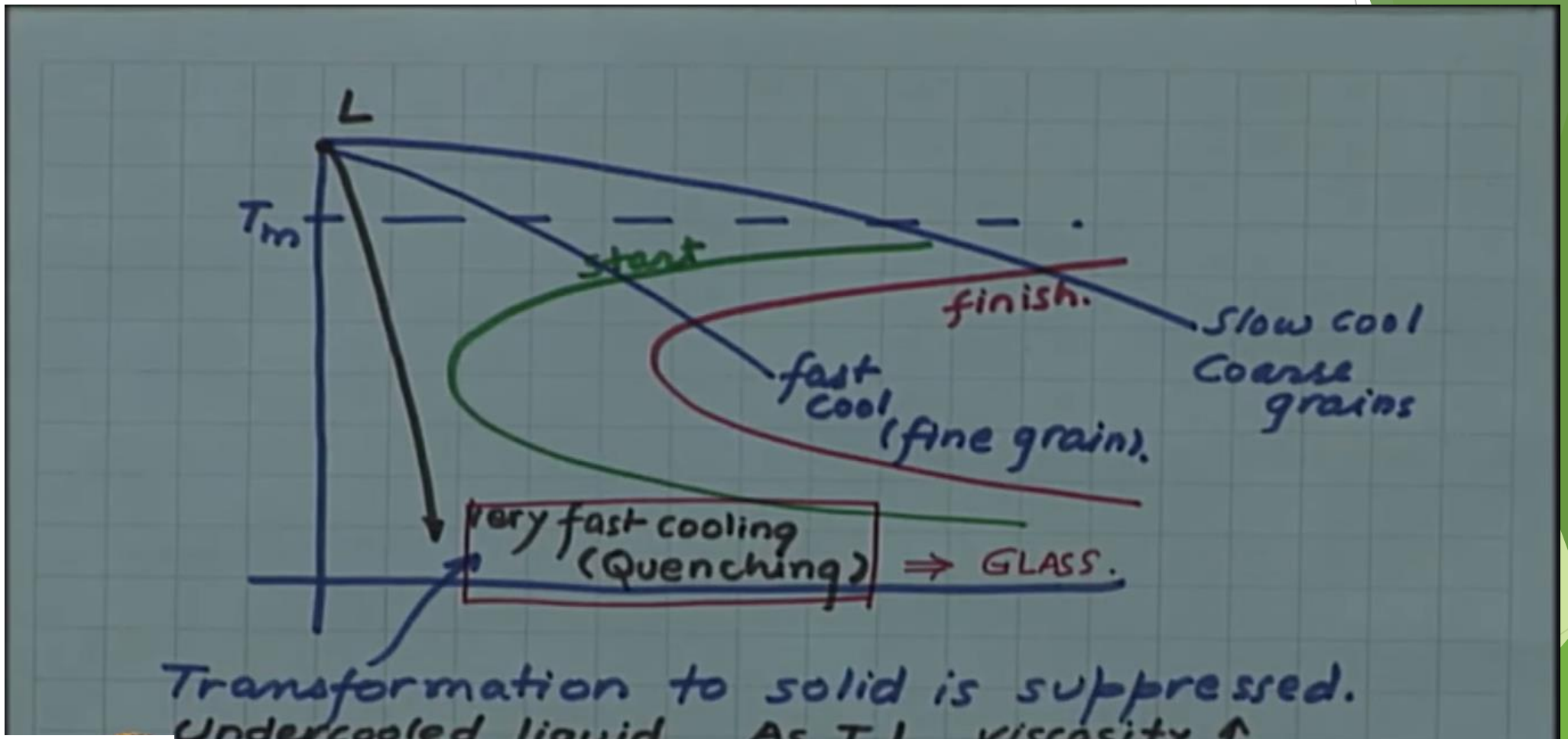
where *k* and *n* are time-independent constants for the particular reaction. The above expression is often referred to as the ***Avrami equation***.

TTT Diagrams (Time-Temp-Transformation)





Can refer to slide 27 again for understanding why coarse and fine grains



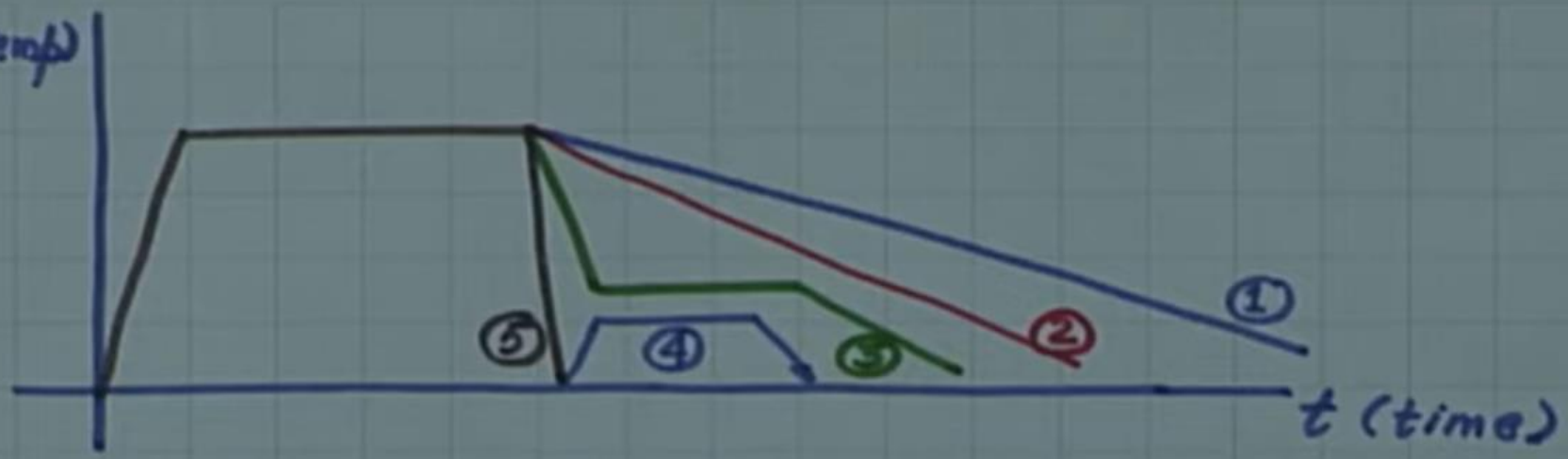
Transformation to solid is suppressed.
Undercooled liquid. As T↓ viscosity ↑
At very low T, viscosity has value like that of solid.
⇒ GLASS or AMORPHOUS SOLID

Heat Treatment Summary

Steel	Hardness HRC	wt% C	Microstructure	Heat Treatment
A	15	0·8	Coarse Pearlite	Annealing
B	30	0·8	Fine Pearlite	Normalizing
C	45	0·8	Bainite	Austempering
D	55	0·8	Tempered Martensite	Tempering
E	65	0·8	Martensite	Quenching

Heat Treatment

Heating a material to a temperature, holiding it at that temperature for a period of time followed by cooling at a specified rate is called heat treatment.

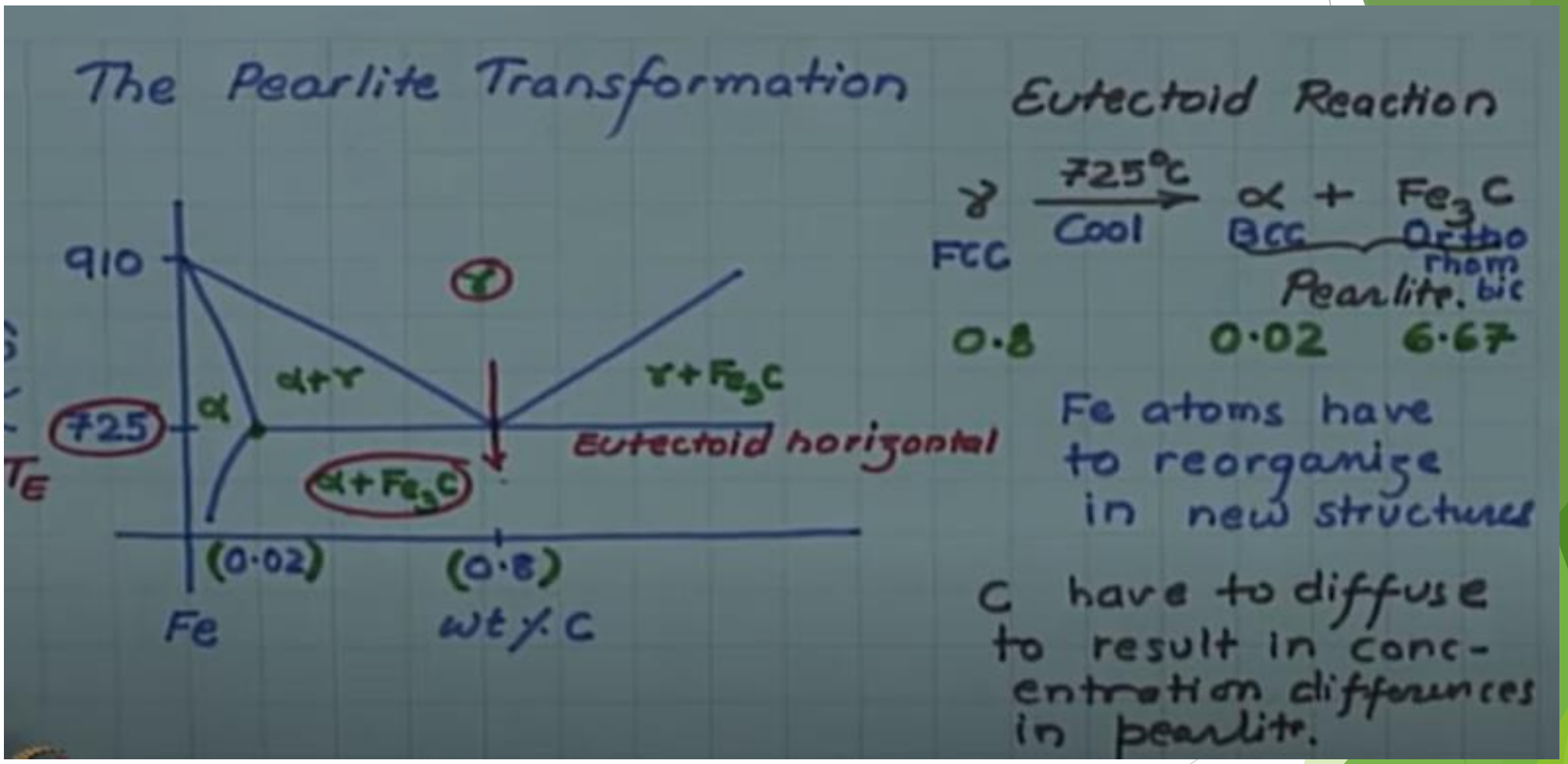


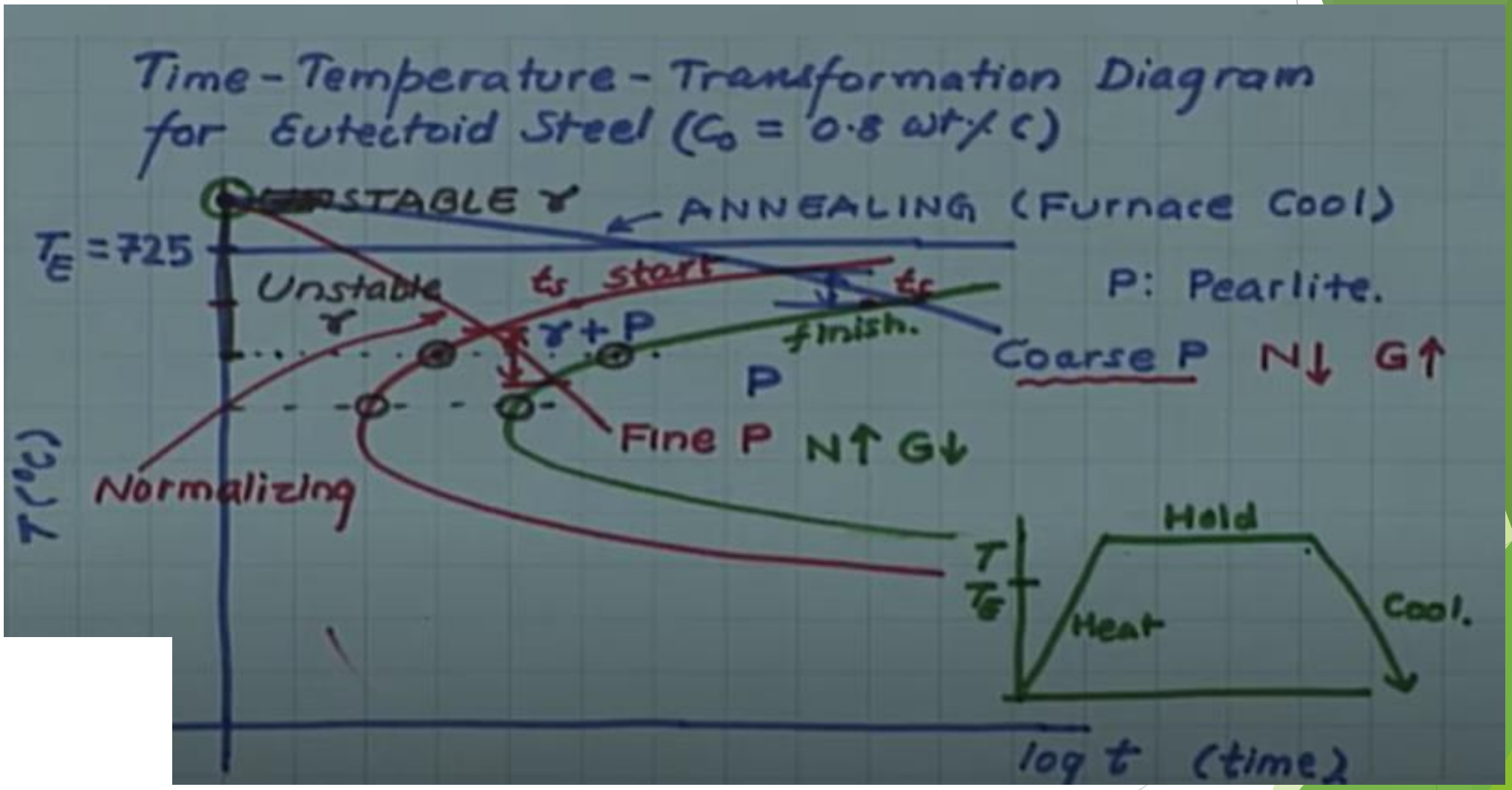
1	Annealing	Slow cooling (Furnace ")	Coarse Pearlite
2.	Normalizing	Faster cooling (Air cooling)	Fine Pearlite
3.	Austempering	Interrupted Quench	Bainite.
	Tempering	Heating after Quench	Tempered Martensite.
	Quenching	V. Fast Cooling (Water)	Martensite

Eqbm Cooling

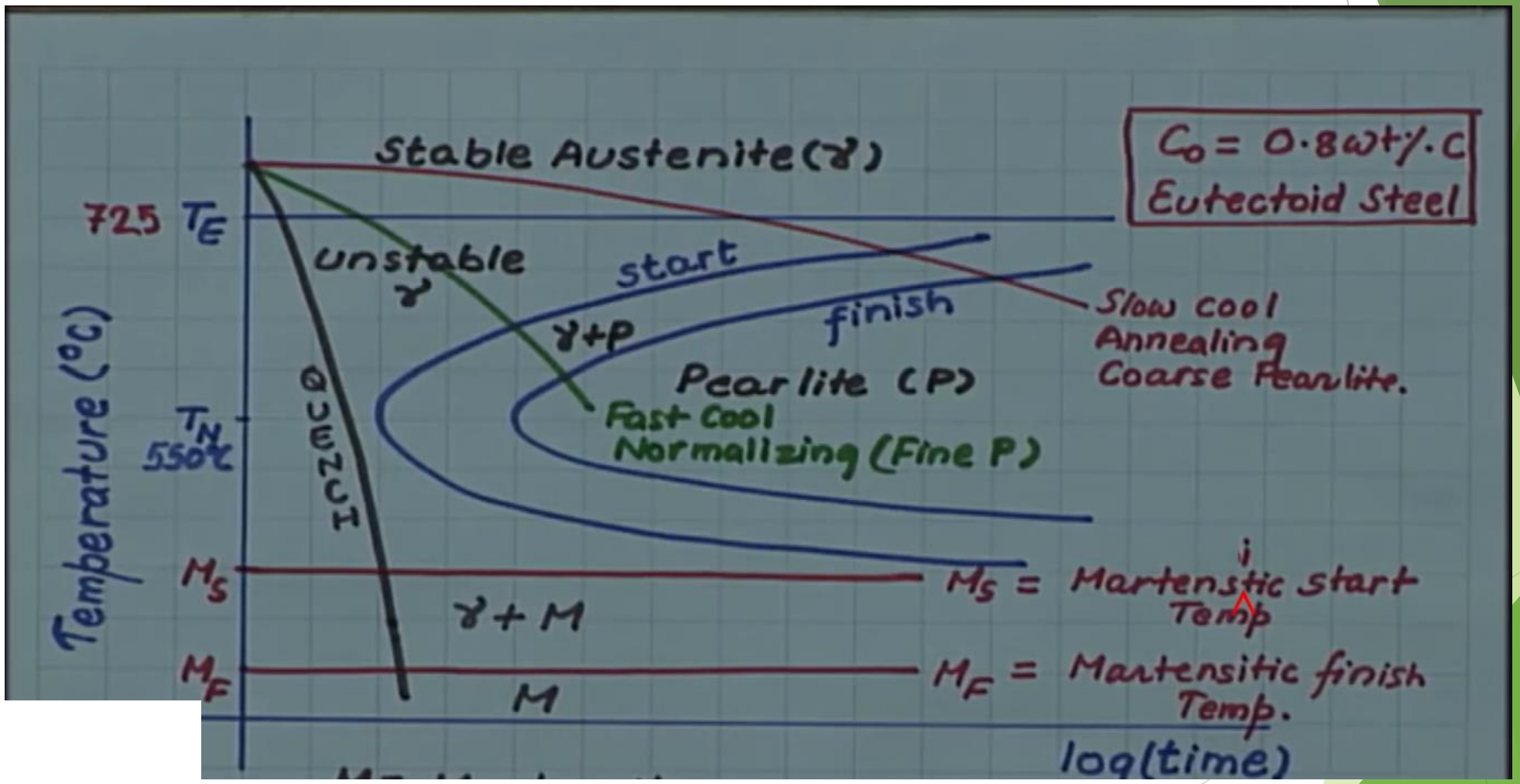
Non Eqbm Cooling

Pearlite

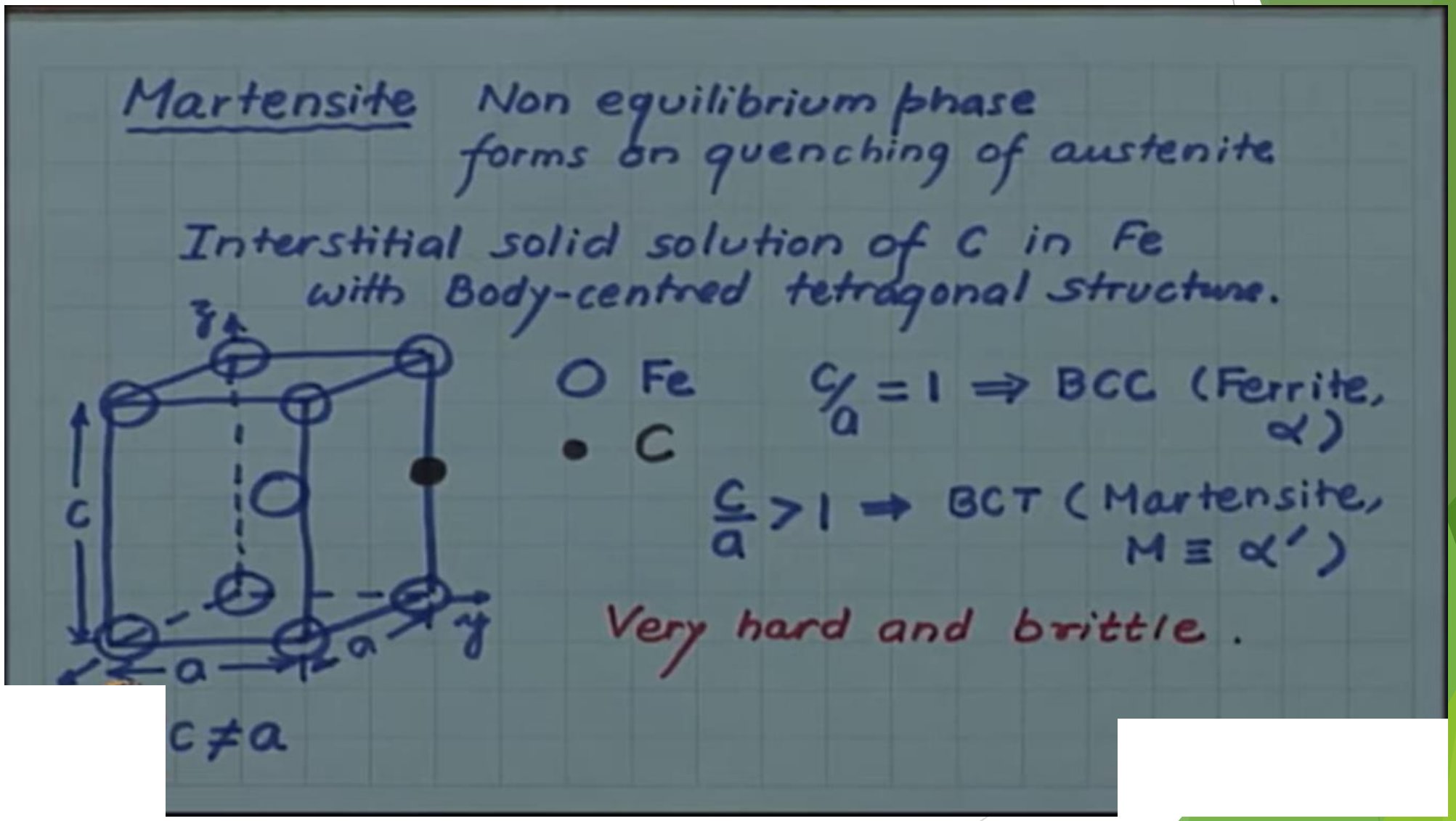

The Pearlite Transformation
910
725
T_E
γ
$\alpha+\gamma$
$\gamma+Fe_3C$
α
$\alpha+Fe_3C$
Eutectoid horizontal
(0.02)
(0.8)
Fe
wt % C
Eutectoid Reaction
$\gamma \xrightarrow[\text{Cool}]{725°C} \alpha + Fe_3C$
FCC
BCC
Ortho rhom bic
Pearlite.
0.8
0.02
6.67
Fe atoms have to reorganize in new structures
C have to diffuse to result in conc-entration differences in pearlite.


Time-Temperature-Transformation Diagram
for Eutectoid Steel (C_0 = 0.8 wt% C)
STABLE γ
ANNEALING (Furnace Cool)
T_E = 725
Unstable γ
t_s start
t_f
finish.
γ + P
P: Pearlite.
Coarse P N↓ G↑
P
Fine P N↑ G↓
Normalizing
T (°C)
Hold
T
T_E
Heat
Cool.
log t (time)

Martensite


Stable Austenite (γ)
C_o = 0.8 wt% C
Eutectoid Steel
725 T_E
unstable γ
start
finish
γ+P
Pearlite (P)
Slow cool
Annealing
Coarse Pearlite.
Fast Cool
Normalizing (Fine P)
T_N
550°C
Temperature (°C)
M_S
γ+M
M_S = Martensitic start Temp
M_F
M
M_F = Martensitic finish Temp.
log(time)

Structure of Martensite



Martensitic Transformation

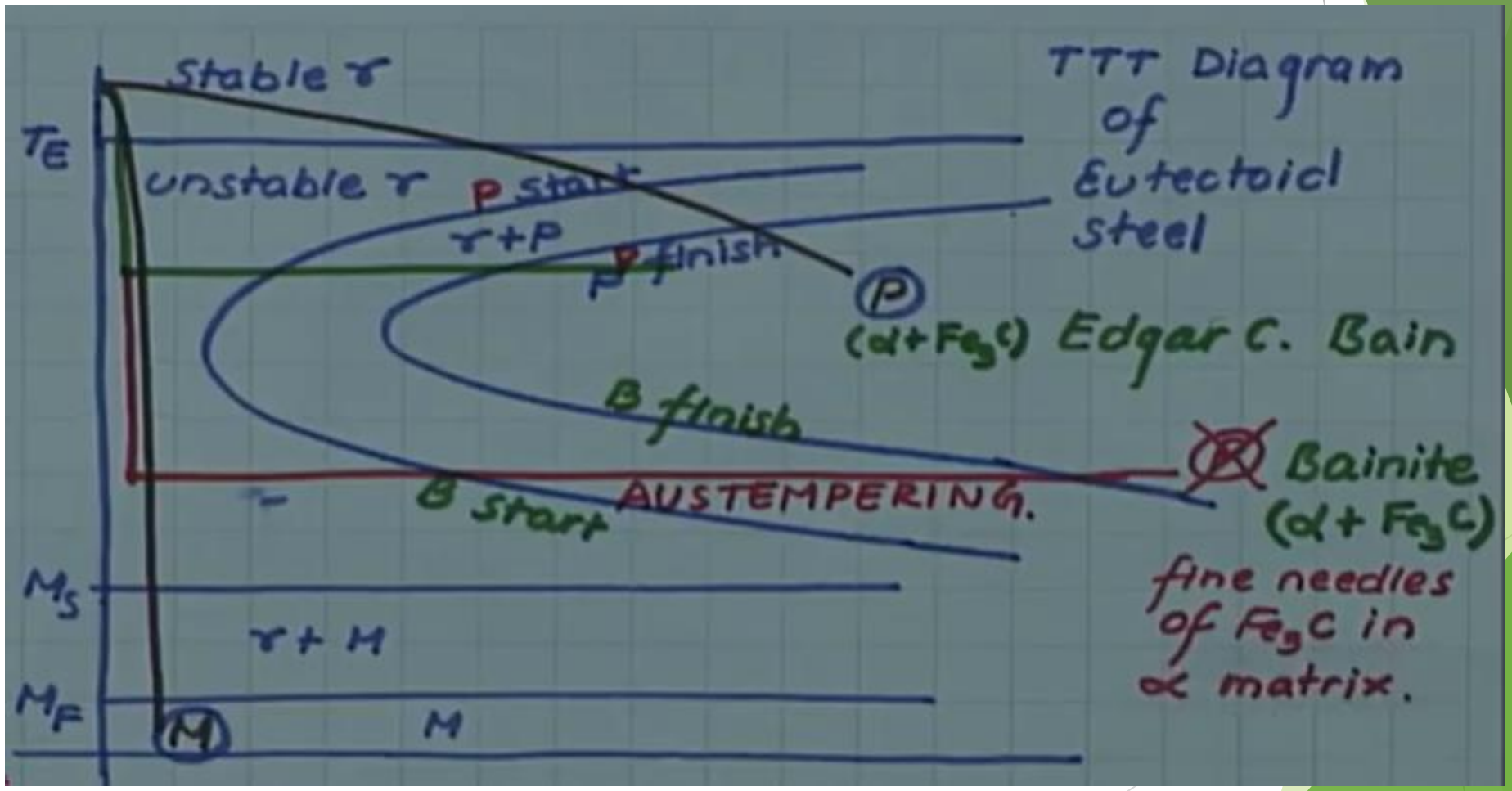
Austenite ⟶ Martensite

1. Transformation is ~~experi~~ extremely rapid.
2. Amount transformed depends only upon temperature and not on time.

   ATHERMAL : Rate of transformation does not depend on temp.
3. No change in composition.

   DIFFUSIONLESS

Bainite


Stable γ
TTT Diagram of Eutectoid steel
T_E
unstable γ
P start
γ+P
P finish
P
(α+Fe_3C) Edgar C. Bain
B finish
B
Bainite
(α+Fe_3C)
AUSTEMPERING.
B start
fine needles of Fe_3C in α matrix.
M_S
γ+M
M_F
M
M

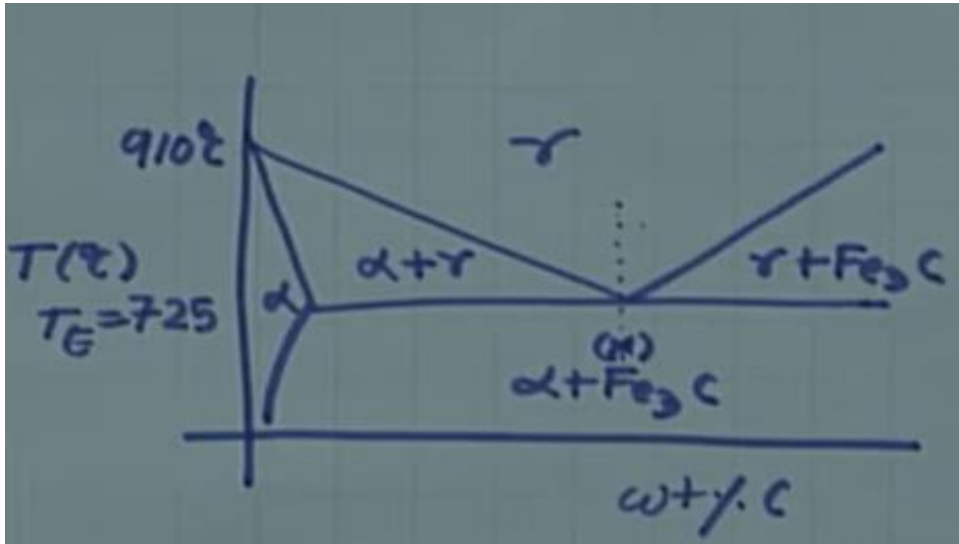
Phases vs Microconstituent.

Pearlite and Bainite are mixtures of the same two phases ($\alpha + Fe_3C$) but are different microconstituents due to different shapes and distribution of these phases.

Tempering

Tempering produces tempered martensite with lower hardness than martensite but better toughness and ductility.

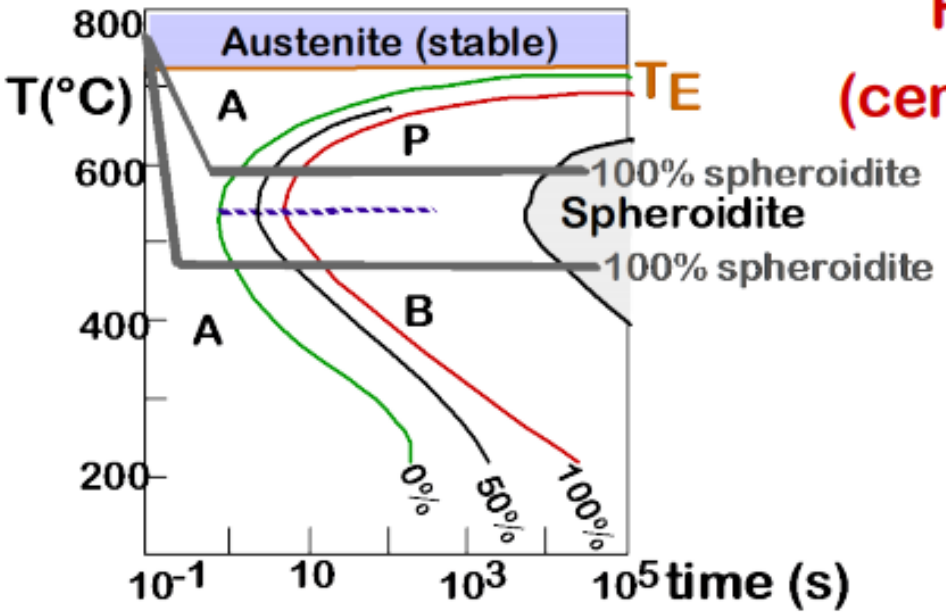
Tempering temp and time control the reduction in hardness and improvement in ductility.

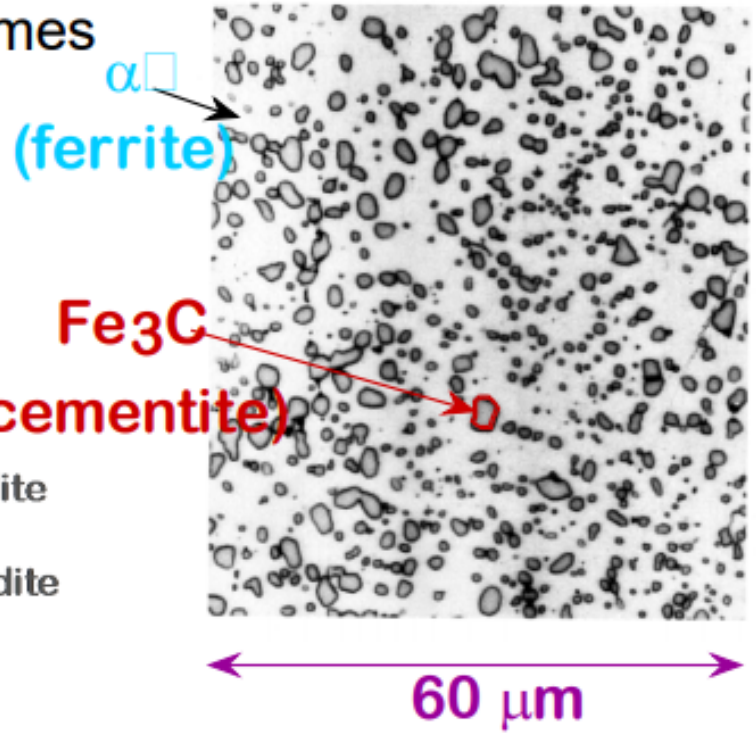


Tempered Martensite
: Fine particles of Fe_3C in α matrix.

$$\gamma \xrightarrow{\text{Quench}} \underset{\text{Hard brittle}}{M} \xrightarrow[\text{Temp, Time}]{\text{Temper}} \underbrace{\alpha + Fe_3C}_{\text{Not pearlite. Tempered Martensite.}}$$

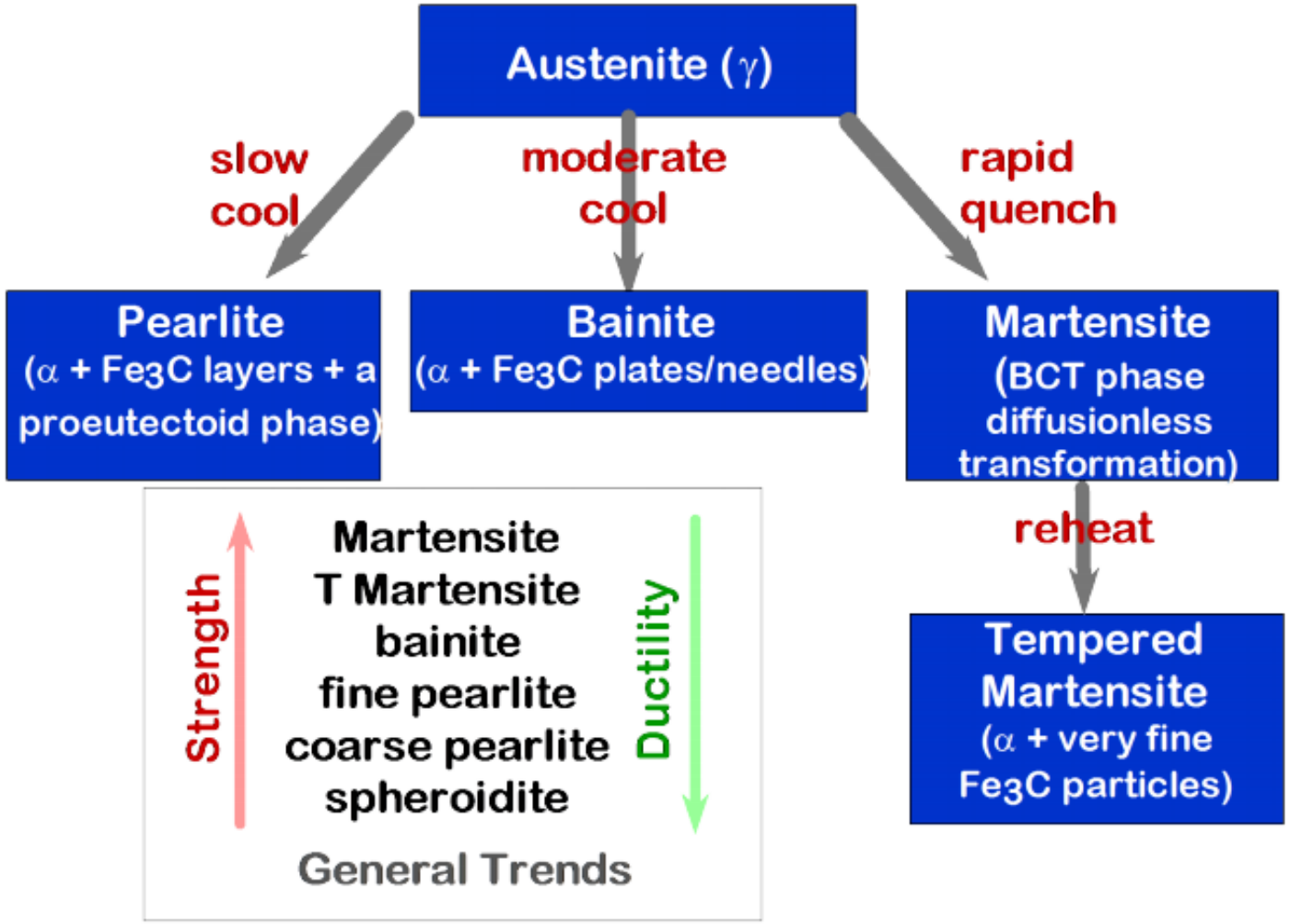
- Spheroidite:
  --a crystals with spherical Fe_3C
  --diffusion dependent.
  --heat bainite or pearlite for long times
  --reduces interfacial area (driving force)
- Isothermal Transf. Diagram





Shift to Sir's slide for CCT

Summary: Processing Options



Go through Mech Prop from Sir's slide

References

- Prof. Grewal’s Slide, SNU
- IMSE, YT